```
> restart;#Flux Phonon;
> eq1 := diff(P(r, t), t $ 2) + V*diff(P(r, t)*V, r $ 2) - V^2*P(r,
  t)/R = 0;# With the flux (-V^2*P(r, t)/R) like the electron color
  Blue
```

$$eq1 := \frac{\partial^2}{\partial t^2} P(r,t) + V^2 \left(\frac{\partial^2}{\partial r^2} P(r,t) \right) - \frac{V^2 P(r,t)}{R} = 0 \tag{1}$$

```
> eq2 := diff(P(r, t), t) + diff(P(r, t)*V, r) = 0;
```

$$eq2 := \frac{\partial}{\partial t} P(r,t) + \left(\frac{\partial}{\partial r} P(r,t) \right) V = 0 \tag{2}$$

```
> iv1 := P(R, 0) = 0, D[1](P)(0, 0) = -R;
```

$$iv1 := P(R,0) = 0, \mathrm{D}_1(P)(0,0) = -R \tag{3}$$

```
> Sol := pdsolve(eq1, HINT = `*`);
```

$$Sol := P(r,t) = f_1(r) f_2(t) \text{ where } \left[\left\{ \frac{\mathrm{d}^2}{\mathrm{d}r^2} f_1(r) = _c_1 f_1(r), \frac{\mathrm{d}^2}{\mathrm{d}t^2} f_2(t) = -V^2 _c_1 f_2(t) + \frac{V^2 f_2(t)}{R} \right\} \right] \tag{4}$$

```
> Sol1 := pdsolve([eq1, eq2, iv1]);
```

$$Sol1 := P(r,t) = \frac{\sqrt{2}\, R^{3/2} \left(e^{\frac{(Vt+2R-r)\sqrt{2}}{2\sqrt{R}}} - e^{-\frac{\sqrt{2}(Vt-r)}{2\sqrt{R}}} \right)}{e^{\sqrt{2}\sqrt{R}} + 1} \tag{5}$$

```
> Flux := Sol1/(4*Pi*r^2);
```

$$Flux := \frac{P(r,t)}{4\pi r^2} = \frac{\sqrt{2}\, R^{3/2} \left(e^{\frac{(Vt+2R-r)\sqrt{2}}{2\sqrt{R}}} - e^{-\frac{\sqrt{2}(Vt-r)}{2\sqrt{R}}} \right)}{4\pi r^2 \left(e^{\sqrt{2}\sqrt{R}} + 1 \right)} \tag{6}$$

```
> F1 := subs(t = 0, Flux);F1 :=R^(3/2)*sqrt(2)*(exp(sqrt(2)*(R - r)
  /sqrt(R)) - 1)*exp(sqrt(2)*r/(2*sqrt(R)))/(4*Pi*r^2*(exp(sqrt(2)*
  sqrt(R)) + 1));
```

$$F1 := \frac{P(r,0)}{4\pi r^2} = \frac{\sqrt{2}\, R^{3/2} \left(e^{\frac{(2R-r)\sqrt{2}}{2\sqrt{R}}} - e^{\frac{\sqrt{2} r}{2\sqrt{R}}} \right)}{4\pi r^2 \left(e^{\sqrt{2}\sqrt{R}} + 1 \right)}$$

$$F1 := \frac{R^{3/2} \sqrt{2} \left(e^{\frac{\sqrt{2}(R-r)}{\sqrt{R}}} - 1 \right) e^{\frac{\sqrt{2} r}{2\sqrt{R}}}}{4\pi r^2 \left(e^{\sqrt{2}\sqrt{R}} + 1 \right)} \tag{7}$$

```
> F:=sqrt(2)*R^(3/2)*sin(sqrt(2)*(R - r)/(2*sqrt(R)))/(4*Pi*r^2*cos
  (sqrt(2)*sqrt(R)/2));# With the flux (+V^2*P(r, t)/R) like the
  Proton with (+)in the Eqution eq1 color Green
```

$$F := \frac{\sqrt{2}\, R^{3/2} \sin\left(\frac{\sqrt{2}\,(R-r)}{2\sqrt{R}}\right)}{4\pi r^2 \cos\left(\frac{\sqrt{2}\,\sqrt{R}}{2}\right)} \tag{8}$$

```
> R:=3.566*10^22; rb:=5.29*10^(-11);plot([F,F1],r=0..9, color =
  ["Green", "Blue"]);
```

$$R := 3.566000000 \times 10^{22}$$

$$rb := 5.290000000 \times 10^{-11}$$

